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ELECTRONIC DEVICE

Abstract

Disclosed is an electronic device including a first body, a second body, and a movable door. The first body has a first pivot side and a heat dissipation opening located on the first pivot side, and the second body has a second pivot side pivoted to the first pivot side. An airflow channel is disposed on the second pivot side, corresponding to the heat dissipation opening. The movable door is rotatably disposed in the airflow channel. When the second body is folded on the first body, the movable door closes the airflow channel. When the second body is unfolded relative to the first body, the movable door rotates relative to the second body to open the airflow channel.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the priority benefit of Taiwan application serial no. 113108031, filed on Mar. 6, 2024. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

[0002] The disclosure relates to an electronic device, and in particular to an electronic device with a heat dissipation design.

Description of Related Art

[0003] Notebook computers have become a type of indispensable electronic devices at work or in daily life for modern people due to their portability, high computing performance, and other features. Specifically, a notebook computer consists of a first body and a second body that are pivoted to each other. The first body is a host capable of logical computing, and the second body is a monitor capable of image display.

[0004] The first body has internal heat sources, including a central processing unit, a graphics processing unit, or other electronic elements that generate a large amount of heat during operation. To quickly discharge the heat generated by the heat source from the inside of the first body to the external environment, an air-cooled heat dissipation module is usually disposed in the first body, corresponding to the heat source. Generally, an air-cooled heat dissipation module includes a fan, a heat pipe, and a heat sink. The fan mostly blows airflow to the rear side of the first body so that hot air is discharged to the external environment through the heat dissipation opening on the rear side of the first body. However, when the second body is unfolded relative to the first body, a lower edge of the second body easily blocks the discharge path of the hot air, causing the hot air to flow to a top surface of the first body (e.g., a surface where a keyboard set and a touch module are disposed) or a display surface of the second body. This results in an increase in the surface temperature of the first body or the second body and even affects the user experience in operation.

SUMMARY

[0005] The disclosure provides an electronic device, which is capable of improving hot air reflux.

[0006] In an embodiment of the disclosure, an electronic device including a first body, a second body, a first magnetic member, and a movable door is provided. The first body has a first pivot side and a heat dissipation opening located on the first pivot side. The first magnetic member is disposed in the first body, corresponding to the first pivot side. The second body has a second pivot side pivoted to the first pivot side. An airflow channel is disposed on the second pivot side, corresponding to the heat dissipation opening. The movable door is rotatably disposed in the airflow channel. A second magnetic member is disposed corresponding to the first magnetic member. When the second body is folded on the first body, a magnetic attraction force generated between the second magnetic member and the first magnetic member fixes the movable door, enabling the movable door to close the airflow channel. When the second body is unfolded relative to the first body, the magnetic attraction force generated between the second magnetic member and the first magnetic member weakens, enabling the movable door to rotate relative to the second body by gravity to open the airflow channel.

[0007] In another embodiment of the disclosure, an electronic device including a first body, a second body, a first magnetic member, a movable door, and a torsion spring is provided. The first body has a first pivot side and a heat dissipation opening located on the first pivot side. The first magnetic member is disposed in the first body, corresponding to the first pivot side. The second body has a second pivot side pivoted to the first pivot side. An airflow channel is disposed on the second pivot side, corresponding to the heat dissipation opening. The movable door is rotatably disposed in the airflow channel. A second magnetic member is disposed on the movable door, corresponding to the first magnetic member. The torsion spring is connected between the movable door and the second body. When the second body is folded on the first body, a magnetic attraction force generated between the second magnetic member and the first magnetic member is greater than an elastic restoring force of the compressed torsion spring, thereby fixing the movable door and enabling the movable door to close the airflow channel. When the second body is unfolded relative to the first body, the magnetic attraction force generated between the second magnetic member and the first magnetic member is smaller than the elastic restoring force of the compressed torsion spring, enabling the movable door to be driven by the torsion spring and rotate relative to the second body to open the airflow channel.

[0008] In still another embodiment of the disclosure, an electronic device including a first body, a second body, and a movable door is provided. The first body has a first pivot side and a heat dissipation opening located on the first pivot side, and the second body has a second pivot side pivoted to the first pivot side. An airflow channel is disposed on the second pivot side, corresponding to the heat dissipation opening. The movable door is rotatably disposed in the airflow channel. When the second body is folded on the first body, the movable door closes the airflow channel. When the second body is unfolded relative to the first body, the movable door rotates relative to the second body to open the airflow channel.

[0009] Based on the above, in the electronic device of the disclosure, when the second body is unfolded relative to the first body, the movable door on the second pivot side of the second body may rotate to open the airflow channel, enabling the hot air in the first body to be discharged through the heat dissipation opening and further flows through the airflow channel to be discharged to an external environment. This prevents the hot air from being blocked by the second body and thus flowing to a top surface of the first body (e.g., a surface where a keyboard set and a touch module are disposed) or a display surface of the second body, which significantly improves hot air reflux.

[0010] To make the aforementioned features and advantages of the disclosure more comprehensible, several embodiments accompanied with drawings are described in detail as follows.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. **1** is a schematic diagram of an electronic device in a first state in an embodiment of the disclosure.

[0012] FIG. **2** is a schematic diagram of the electronic device in FIG. **1** transitioning to a second state.

[0013] FIG. **3** is a schematic diagram of FIG. **2** in another view.

[0014] FIG. **4** is a partially enlarged schematic diagram of a region R**1** in FIG. **1**.

[0015] FIG. **5** is a partially enlarged schematic diagram of a region R**2** in FIG. **2**.

[0016] FIG. **6** is a partially enlarged schematic diagram of a region R**3** in FIG. **3**.

[0017] FIG. **7** is a partial cross-sectional schematic diagram of FIG. **4** along a section line A-A.

[0018] FIG. **8** is a partial cross-sectional schematic diagram of FIG. **7** transitioned to the second state.

DESCRIPTION OF THE EMBODIMENTS

[0019] FIG. **1** is a schematic diagram of an electronic device in a first state in an embodiment of the disclosure. FIG. **2** is a schematic diagram of the electronic device in FIG. **1** transitioning to a second state. FIG. **3** is a schematic diagram of FIG. **2** in another view. Referring to FIGS. **1** to **3**, in this embodiment, an electronic device **100** may be a notebook computer and includes a first body **110** and a second body **120** pivoted to each other. The first body **110** is a host capable of logical computing, and the second body **120** is a monitor capable of image display.

[0020] The second body **120** is pivoted to the first body **110** by a hinge structure **130** to be folded on or unfolded relative to the first body **110**, for example, switching between a folded state (i.e., a first state) and an unfolded state (i.e., a second state). Further, the first body **110** has a first pivot side **111**, wherein the second body **120** has a second pivot side **121** pivoted to the first pivot side **111**, and the hinge structure **130** is connected between the second pivot side **121** and the first pivot side **111**.

[0021] FIG. **4** is a partially enlarged schematic diagram of a region R**1** in FIG. **1**. FIG. **5** is a partially enlarged schematic diagram of a region R**2** in FIG. **2**. FIG. **6** is a partially enlarged schematic diagram of a region R**3** in FIG. **3**. Referring to FIGS. **1** and **4**, in this embodiment, the first body **110** also has a heat dissipation opening **112** located on the first pivot side **111**. An airflow channel **122** is disposed on the second pivot side **121**, corresponding to the heat dissipation opening **112**. Specifically, the electronic device **100** further includes a movable door **140**, wherein the movable door **140** is pivoted to the second pivot side **121** and rotatably disposed in the airflow channel **122**.

[0022] In the first state, the second body **120** is folded on the first body **110**, and the entire movable door **140** is accommodated in the airflow channel **122** to close the airflow channel **122**, as shown in FIGS. **1** and **4**. In the second state, the second body **120** is unfolded relative to the first body **110**, and at least a part of the movable door **140** moves out of the airflow channel **122** to open the airflow channel **122**, as shown in FIGS. **2** and **5** or FIGS. **3** and **6**.

[0023] For example, the airflow channel **122** is a groove located on the second pivot side **121**, and the entire movable door **140** is adaptable to be accommodated in the groove. As the second body **120** is unfolded relative to the first body **110**, the movable door **140** rotates relative to the second pivot side **121**, enabling at least a part of the movable door **140** to move out of the groove.

[0024] FIG. **7** is a partial cross-sectional schematic diagram of FIG. **4** along a section line A-A. FIG. **8** is a partial cross-sectional schematic diagram of FIG. **7** transitioned to the second state. Referring to FIGS. **4** to **7**, the second body **120** has a display surface **123** and a back surface **124** relative to the display surface **123**. The airflow channel **122** penetrates the display surface **123** and the back surface **124** to form an air inlet opening **122***a* and an air exhaust opening **122***b* on the display surface **123** and the back surface **124**, respectively. On the other hand, the first body **110** has a top surface **113** and a bottom surface **114** relative to the top surface **113**. In the first state, the second body **120** is folded on the first body **110**, wherein the display surface **123** is bonded to the top surface **113**, and the entire movable door **140** is accommodated in the airflow channel **122** to close the air inlet opening **122***a* and the air exhaust opening **122***b*.

[0025] As shown in FIGS. **5**, **6**, and **8**, in the second state, the second body **120** is unfolded relative to the first body **110**. The movable door **140** rotates relative to the second pivot side **121**, enabling at least a part of the movable door **140** to move out of the airflow channel **122** through the air exhaust opening **122***b* to open the air inlet opening **122***a* and the air exhaust opening **122***b*.

[0026] Opening the airflow channel **122** or opening the air inlet opening **122***a* and the air exhaust opening **122***b* as previously mentioned refers to that the airflow is not blocked by the movable door **140** and flows to the air exhaust opening **122***b* through the airflow channel **122** or from the air inlet opening **122***a*. Relatively, closing the airflow channel **122** or closing the air inlet opening **122***a* and the air exhaust opening **122***b* as previously mentioned refers to that the airflow is blocked by the movable door **140** and unable to flow to the air exhaust opening **122***b* through the airflow channel **122** or from the air inlet opening **122***a*.

[0027] As shown in FIGS. **7** and **8**, in an embodiment, the movable door **140** may be used to close the airflow channel **122** or open the airflow channel **122** by the gravity of the movable door **140**. When the second body **120** is folded on the first body **110**, the display surface **123** is bonded to the top surface **113**, and the movable door **140** gets close to the top surface **113** by the gravity of the movable door **140** to close the airflow channel **122**.

[0028] When the second body **120** is unfolded relative to the first body **110**, the display surface **123** separates from the top surface **113**, and the movable door **140** rotates relative to the second pivot side **121** by the gravity of the movable door **140**, enabling at least a part of the movable door **140** to move out of the airflow channel **122** through the air exhaust opening **122***b* to open the airflow channel **122**.

[0029] As shown in FIGS. **7** and **8**, in another embodiment, the electronic device **100** further includes a first magnetic member **150**, wherein the first magnetic member **150** is disposed in the first body **110**, corresponding to the first pivot side **111**. A second magnetic member **160** is disposed on the movable door **140**, corresponding to the first magnetic member **150**. Specifically, the first magnetic member **150** and the second magnetic member **160** may be two magnets or a combination of a magnet and an attracted body, which are adaptable to generate a magnetic attraction force between each other. When the second body **120** is folded on the first body **110**, the magnetic attraction force generated between the second magnetic member **160** and the first magnetic member **150** fixes the movable door **140**, enabling the movable door **140** to close the airflow channel **122** to prevent the movable door **140** from accidentally moving out of the airflow channel **122**.

[0030] When the second body **120** is unfolded relative to the first body **110**, the distance between the second magnetic member **160** and the first magnetic member **150** increases as the second body **120** is unfolded relative to the first body **110**, enabling the magnetic attraction force generated between the second magnetic member **160** and the first magnetic member **150** to weaken. When the magnetic attraction force generated between the second magnetic member **160** and the first magnetic member **150** weakens and becomes smaller than the gravity of the movable door **140**, the movable door **140** rotates relative to the second pivot side **121** by the gravity of the movable door **140**, enabling at least a part of the movable door **140** to move out of the airflow channel **122** through the air exhaust opening **122***b* to open the airflow channel **122**.

[0031] As shown in FIGS. **7** and **8**, in still another embodiment, the electronic device **100** further includes a torsion spring **170** connected between the movable door **140** and the second body **120**. When the second body **120** is folded on the first body **110**, the magnetic attraction force generated between the second magnetic member **160** and the first magnetic member **150** fixes the movable door **140**, enabling the movable door **140** to close the airflow channel **122** to prevent the movable door **140** from accidentally moving out of the airflow channel **122**. Further, in a state where the movable door **140** closes the airflow channel **122**, the torsion spring **170** is compressed by the movable door **140** and the elastic restoring force of the torsion spring **170** is smaller than the magnetic attraction force generated between the second magnetic member **160** and the first magnetic member **150**, enabling the movable door **140** to be fixed in the airflow channel **122** by the magnetic attraction force generated between the second magnetic member **160** and the first magnetic member **150**.

[0032] When the second body **120** is unfolded relative to the first body **110**, the distance between the second magnetic member **160** and the first magnetic member **150** increases as the second body **120** is unfolded relative to the first body **110**, enabling the magnetic attraction force generated between the second magnetic member **160** and the first magnetic member **150** to weaken. When the magnetic attraction force generated between the second magnetic member **160** and the first magnetic member **150** weakens and becomes smaller than the elastic restoring force of the torsion spring **170**, the movable door **140** is driven by the torsion spring **170** and rotates relative to the second pivot side **121** by the gravity of the movable door **140**, enabling at least a part of the movable door **140** to move out of the airflow channel **122** through the air exhaust opening **122***b* to open the airflow channel **122**.

[0033] As shown in FIGS. **7** and **8**, in this embodiment, the first magnetic member **150** is adjacent to the top surface **113**. That is, the distance between the first magnetic member **150** and the top surface **113** is smaller than the distance between the first magnetic member **150** and the bottom surface **114**. On the other hand, the electronic device **100** further includes a fan **180** disposed in the first body **110**, wherein the first magnetic member **150** is located between the fan **180** and the heat dissipation opening **112**. Moreover, the fan **180** may be a centrifugal fan for blowing cool air toward the heat dissipation opening **112**.

[0034] As shown in FIG. **8**, a hot air **101** in the first body **110** is adapted to flow toward the heat dissipation opening **112** and be discharged through the heat dissipation opening **112**. In the second state, the airflow channel **122** is opened, enabling the hot air **101** to further flow through the airflow channel **122** to be discharged to the external environment after being discharged through the heat dissipation opening **112**. This prevents the hot air **101** from being blocked by the second body **120** and thus flowing to the top surface **113** of the first body **110** (e.g., a surface where a keyboard set and a touch module are disposed) or the display surface **123** of the second body **120**, which significantly improves the reflux of the hot air **101**. With the reflux of the hot air **101** being improved, the surface temperature of the first body **110** or the second body **120** is less likely to increase under the impact of the hot air **101**, thereby optimizing the user experience in operation.

[0035] As shown in FIG. **8**, in the second state, the hot air **101** may be further blown toward the air inlet opening **122***a* after being discharged through the heat dissipation opening **112**. Then, the hot air **101** may flow from the air inlet opening **122***a* to the air exhaust opening **122***b* to be discharged to the external environment, for example, to the surrounding area on a rear side **124** of the second body **120**, through the air exhaust opening **122***b*.

[0036] In summary, in the electronic device of the disclosure, when the second body is unfolded relative to the first body, the movable door on the second pivot side of the second body may rotate to open the airflow channel, enabling the hot air in the first body to be discharged through the heat dissipation opening and further flows through the airflow channel to be discharged to an external environment. This prevents the hot air from being blocked by the second body and thus flowing to the top surface of the first body (e.g., the surface where the keyboard set and the touch module are disposed) or the display surface of the second body, which significantly improves hot air reflux. With hot air reflux being improved, the surface temperature of the first body or the second body is less likely to increase under the impact of hot air, thereby optimizing the user experience in operation.

[0037] Although the disclosure has been disclosed in the above embodiments, the embodiments are not intended to limit the disclosure. Persons skilled in the art may make some changes and modifications without departing from the spirit and scope of the disclosure. Therefore, the protection scope of the disclosure shall be defined by the appended claims.

Claims

1. An electronic device, comprising: a first body, having a first pivot side and a heat dissipation opening located on the first pivot side; a second body, having a second pivot side pivoted to the first pivot side, wherein an airflow channel is disposed on the second pivot side, the airflow channel corresponding to the heat dissipation opening; a first magnetic member, corresponding to the first pivot side and disposed in the first body; and a movable door, rotatably disposed in the airflow channel, wherein a second magnetic member is disposed corresponding to the first magnetic member, when the second body is folded on the first body, a magnetic attraction force generated between the second magnetic member and the first magnetic member fixes the movable door, enabling the movable door to close the airflow channel, and when the second body is unfolded relative to the first body, the magnetic attraction force generated between the second magnetic member and the first magnetic member weakens, enabling the movable door to rotate relative to the second body by gravity to open the airflow channel.

2. The electronic device of claim 1, wherein a distance between the second magnetic member and the first magnetic member increases as the second body is unfolded relative to the first body.

3. The electronic device of claim 1, wherein the first body has a top surface and a bottom surface relative to the top surface, and the first magnetic member is adjacent to the top surface.

4. The electronic device of claim 1, further comprising a fan disposed in the first body, wherein the first magnetic member is located between the fan and the heat dissipation opening.

5. The electronic device of claim 1, wherein the airflow channel is a groove located on the second pivot side, and the entire movable door is adaptable to be accommodated in the groove, wherein at least a part of the movable door moves out of the groove as the second body is unfolded relative to the first body.

6. The electronic device of claim 1, wherein the second body has a display surface and a back surface relative to the display surface, and the airflow channel penetrates the display surface and the back surface.

7. The electronic device of claim 6, wherein the airflow channel has an air inlet opening located on the display surface and an air exhaust opening located on the back surface, wherein hot air in the first body is adaptable to be discharged through the heat dissipation opening and blown to the air inlet opening, and then flowing from the air inlet opening to the air exhaust opening to be discharged to an external environment through the air exhaust opening.

8. The electronic device of claim 6, wherein the airflow channel has an air inlet opening located on the display surface and an air exhaust opening located on the back surface, wherein at least a part of the movable door is adaptable to move out of the airflow channel through the air exhaust opening.

9. An electronic device, comprising: a first body, having a first pivot side and a heat dissipation opening located on the first pivot side; a second body, having a second pivot side pivoted to the first pivot side, wherein an airflow channel is disposed on the second pivot side, the airflow channel corresponding to the heat dissipation opening; a first magnetic member, corresponding to the first pivot side and disposed in the first body; a movable door, rotatably disposed in the airflow channel, wherein a second magnetic member is disposed corresponding to the first magnetic member; and a torsion spring, connected between the movable door and the second body, when the second body is folded on the first body, a magnetic attraction force generated between the second magnetic member and the first magnetic member is greater than an elastic restoring force of the compressed torsion spring, thereby fixing the movable door and enabling the movable door to close the airflow channel, and when the second body is unfolded relative to the first body, the magnetic attraction force generated between the second magnetic member and the first magnetic member is smaller than the elastic restoring force of the torsion spring, enabling the movable door to be driven by the torsion spring and rotate relative to the second body to open the airflow channel.

10. An electronic device, comprising: a first body, having a first pivot side and a heat dissipation opening located on the first pivot side; a second body, having a second pivot side pivoted to the first pivot side, wherein an airflow channel is disposed on the second pivot side, the airflow channel corresponding to the heat dissipation opening; and a movable door, rotatably disposed in the airflow channel, when the second body is folded on the first body, the movable door closes the airflow channel, and when the second body is unfolded relative to the first body, the movable door rotates relative to the second body to open the airflow channel.